

Question Number	Answer	Additional Guidance	Mark
1(a)(i)	{cristae / inner (mitochondrial) membrane}	Accept stalked particles	(1) Grad

Question Number	Answer	Additional Guidance	Mark
1(a)(ii)	NAD / FAD	Accept cytochrome oxidase / cytochromes / NADH / FADH ₂ Ignore reference to oxidised and reduced Do not accept NADP or other incorrect molecule No mark if one correct and one incorrect answer given	(1) exp

Question Number	Answer	Additional Guidance	Mark
1(b)	A description that contains five of the following: • range of five suitable temperatures used (1) • use of thermostatic waterbath (1) • incubate yeast and indicator separately to reach same temperature (1) • suitable named indicator used (1) • record time taken for colour to change (1) • identification of one variable (1) • repeats and calculate { means/SD's } (to compare) (1)	Accept temperatures between 5 and 45°C (ignore additional temperatures outside the range) Accept waterbath set at a stated temperature Accept to equilibrate/acclimatise Accept TTC / DCPIP / methylene blue ignore stated colours Accept: pH {volume / conc/mass/strain/type} of yeast suspension/ volume/conc of TTC Ignore species Accept average	Exp (5)

Question Number	Answer	Additional Guidance	Mark
1(c)	An explanation that includes three of the following: • at high temperatures there is more kinetic energy (1) • as temperature increases { more collisions between enzyme and substrate/more ES (complexes) formed} (1) • at {high(er) temperatures /above optimum temperature} enzymes are denatured (1)	Accept reverse argument / KE Accept reverse argument Accept description of bonds being broken changing shape of active site Accept high temperatures change shape of active site so substrate no longer fits not denatured if low temperature	Exp (3)

(Total for Question 1 = 10 marks)

